PROJECT NOTES

GENERAL

1. THESE DRAWINGS SHALL BE READ IN CONJUNCTION WITH SPECIFICATIONS AND OTHER CONSULTANT'S DRAWINGS.
2. THE WEATHER PROOFING OF THE BUILDING IS THE ARCHITECT'S/SUBCONTRACTOR'S RESPONSIBILITY. THIS INCLUDES (BUT IS NOT LIMITED TO) THE SPECIFICATION AND FIXING DETAILS OF CLADDINGS, SHEETING, FLASHING, MEMBRANES, STEPS, SET DOWNS & RECESSES.
3. ALL DISCREPANCIES SHALL BE REFERRED TO THE MANAGING CONTRACTOR AND RESOLVED BEFORE PROCEEDING WITH THE WORK.
4. REFER ARCHITECT ALL DIMENSIONS AND SET OUT THAT SHALL BE VERIFIED BY THE SUBCONTRACTOR ON SITE.
5. **DO NOT SCALE STRUCTURAL DRAWINGS.**
6. DIMENSIONS SHOWN ON STRUCTURAL DRAWINGS PROVIDE THE MAXIMUM OR MINIMUM STRUCTURAL DESIGN REQUIREMENTS AND ARE NOT TO BE USED FOR SET OUT. IF ARCHITECTURAL SET OUT EXCEEDS THE STRUCTURAL DESIGN REQUIREMENTS DO NOT PROCEED UNTIL THE DESIGN REQUIREMENTS HAVE BEEN RESOLVED.
7. THE REDUCED LEVELS RL SHOWN ON STRUCTURAL DRAWINGS ARE INDICATIVE AND ARE PROVIDED TO SHOW THE GENERAL DESIGN INTENT. THE CONTRACTOR SHALL CONFIRM ALL LEVELS ON SITE WITH AUSTRALIAN HEIGHT DATA, CIVIL ENGINEERING AND ARCHITECTURAL DRAWINGS AND TRANSLATE STRUCTURAL LEVELS AS REQUIRED WHILST MAINTAINING THE DESIGN INTENT.
8. ALL MEASUREMENTS INDICATED ON STRUCTURAL DRAWINGS ARE IN MILLIMETERS (mm) UNLESS NOTED OTHERWISE.
9. ALL WORKMANSHIP, TESTING, MATERIALS AND SUPERVISION ARE TO BE IN ACCORDANCE WITH THE RELEVANT AUSTRALIAN STANDARDS, THESE SPECIFICATIONS, AND THE WORKPLACE HEALTH AND SAFETY ACT 2011.
10. PROPRIETARY ITEMS SPECIFIED SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S WRITTEN RECOMMENDATIONS. DO NOT VARY SPECIFIED PROPRIETARY PRODUCTS WITHOUT WRITTEN APPROVAL FROM THE ENGINEER.
11. THESE DRAWINGS AND ISSUED WRITTEN INSTRUCTIONS DURING THE COURSE OF THE CONTRACT DEPICT THE COMPLETE STRUCTURE. THEY DO NOT DESCRIBE A WORK METHOD. THE ARRANGEMENT, DESIGN AND INSTALLATION OF TEMPORARY WORKS REMAINS THE RESPONSIBILITY OF THE SUBCONTRACTOR.
12. THE SUBCONTRACTOR SHALL PROVIDE CERTIFICATION ON ANY DESIGN AND CONSTRUCT COMPONENT BY A CHARTERED PROFESSIONAL ENGINEER (RPEQ).
13. THE SUBCONTRACTOR SHALL BE RESPONSIBLE FOR THE LOCATION OF ALL SERVICES IN THE VICINITY OF THE WORKS. ANY SERVICES SHOWN ARE PROVIDED FOR INFORMATION ONLY. THE SUBCONTRACTOR SHALL CONFIRM THE LOCATION OF ALL SERVICES PRIOR TO COMMENCING AND SHALL BE RESPONSIBLE FOR THE REPAIR OF ANY DAMAGE CAUSED TO SERVICES, AS WELL AS ANY LOSS INCURRED AS A RESULT OF THE DAMAGE TO ANY SERVICES.
14. THE METHOD OF CONSTRUCTION AND THE MAINTENANCE OF SAFETY DURING CONSTRUCTION IS THE RESPONSIBILITY OF THE SUBCONTRACTOR. IF ANY STRUCTURAL ELEMENT PRESENTS DIFFICULTY IN RESPECT TO SAFETY THE MATTER SHALL BE REFERRED TO NORTHROP CONSULTING ENGINEERS FOR RESOLUTION BEFORE PROCEEDING WITH THE WORK. REFER WORKPLACE HEALTH AND SAFETY REGISTER
15. CONSTRUCTION USING THESE DRAWINGS SHALL NOT COMMENCE UNTIL A CONSTRUCTION CERTIFICATE HAS BEEN ISSUED AND ONLY IF THE DRAWINGS ARE DESIGNATED ISSUED FOR CONSTRUCTION.
16. NO CHANGES IN ANY STRUCTURAL ELEMENT SHALL BE MADE WITHOUT WRITTEN APPROVAL FROM NORTHROP CONSULTING ENGINEERS. IF THERE IS A DISCREPANCY THEN FOR TENDER PURPOSES ALLOW FOR THE MOST EXPENSIVE OPTION. NORTHROP CONSULTING ENGINEERS SHALL BE CONTACTED TO CONFIRM PRIOR TO CONSTRUCTION.
17. NORTHROP CONDUCTS PERIODIC INSPECTIONS OF THE CONSTRUCTION PROCESS TO ENSURE CONFORMANCE WITH ENGINEERING DOCUMENTATION. IT IS THE CONTRACTORS RESPONSIBILITY TO ENSURE ELEMENTS ARE ERECTED SAFELY AND IN A MANNER THAT WILL NOT AFFECT THE ELEMENTS ABILITY TO FUNCTION TO ITS DESIGN INTENT.
18. NORTHROP CONSULTING ENGINEERS ACCEPTS NO RESPONSIBILITY FOR ANY CONSTRUCTION WORKS.
19. THE STRUCTURAL COMPONENTS DETAILED ON THESE STRUCTURAL DRAWINGS ARE JOB SPECIFIC AND HAVE BEEN DESIGNED IN ACCORDANCE WITH THE RELEVANT AUSTRALIAN STANDARDS AND BUILDING CODE OF AUSTRALIA USING THE FOLLOWING DESIGN PARAMETERS AND LOADING.

WIND LOADS TO AS/NZS 1170.2:2021	
IMPORTANCE LEVEL	2
REGION	B
ANNUAL PROBABILITY OF EXCEEDANCE (ULS)	1:500
REGIONAL WIND SPEED V (m/s)(ULS)	VARIES
TERRAIN CATEGORY	VARIES
TERRAIN MULTIPLIER Mz, CAT	VARIES
WIND DIRECTION MULTIPLIER Md	VARIES
SHIELDING MULTIPLIER Ms	VARIES
TOPOGRAPHIC MULTIPLIER Mt	1
SITE WIND SPEED (m/s)(ULS)	VARIES

EARTHQUAKE LOADS	
IMPORTANCE LEVEL	2
ANNUAL PROBABILITY OF EXCEEDANCE (P)	1:500
PROBABILITY FACTOR kp	1.0
HAZARD FACTOR Z	0.08
SITE SUB-SOIL CLASS	Ce SHALLOW SOIL
EARTHQUAKE DESIGN CATEGORY EDC	II
DESIGN REQUIRED	STATIC ANALYSIS

LIVE LOADS	
FLOORS	REFER RELEVANT DRAWINGS
ROOFS	REFER RELEVANT DRAWINGS

SUPERIMPOSED DEAD LOADS	
FLOORS	REFER RELEVANT DRAWINGS
ROOFS	REFER RELEVANT DRAWINGS

CONCRETE EXPOSURE CLASSIFICATION	
EXTERNAL	B1
INTERNAL	A2

STEELWORK CONSTRUCTION CATEGORY	
SERVICE CATEGORY	SC1
FABRICATION CATEGORY	FC1
CONSTRUCTION CATEGORY	CC2

TEMPORARY WORKS

1. THESE DRAWINGS DEPICT THE "PERMANENT" STRUCTURE, TEMPORARY WORKS REMAIN THE RESPONSIBILITY OF THE SUBCONTRACTOR.
2. SUBCONTRACTOR MUST ENGAGE (RPEQ) QUALIFIED STRUCTURAL ENGINEER FOR THE DESIGN OF ALL TEMPORARY WORKS NECESSARY TO SAFELY ERECT THIS STRUCTURE. AS A MINIMUM THE FOLLOWING WORKS FORMWORK / TEMPORARY PROPPING / NEEDLE BEAMS / SCAFFOLDING / UNDERPINNING
3. SUBCONTRACTOR SHALL CONTACT NORTHROP CONSULTING ENGINEERS IF THEY CONSIDER ANY PART OF THIS STRUCTURE IS UNSAFE TO ERECT

CONCRETE NOTES

GENERAL

1. CARRY OUT ALL CONCRETE WORKS IN ACCORDANCE WITH CURRENT REVISIONS OF AS1379, AS3600, AS3610, AND NATSPEC CONCRETE STANDARDS.
2. ALL CONCRETE TO BE NOMINAL (N) CLASS IN ACCORDANCE WITH AS1379 CLAUSE 1.5.3.2 WITH THE FOLLOWING CONCRETE MIX PARAMETERS:

f'c 28 DAY COMPRESSIVE STRENGTH	REFER RELEVANT DRAWING
SLUMP	SUPPLIER TO RECOMMEND TO SUIT PLACEMENT
AGGREGATE SIZE	20

3. WHERE CONCRETE ON REFERENCE DOCUMENTATION IS NOTED AS 'S' SPECIAL CLASS IT SHALL COMPLY WITH AS1379 AND HAVE THE FOLLOWING CONCRETE MIX PARAMETERS:

f'c 28 DAY COMPRESSIVE STRENGTH	REFER RELEVANT DRAWING
SLUMP	SUPPLIER TO RECOMMEND TO SUIT PLACEMENT
AGGREGATE SIZE	20
MINIMUM CEMENT RATIO	250kg/m^3
NOMINAL 56 DAY SHRINKAGE STRAIN	650um

4. WHERE CONCRETE ON REFERENCE DOCUMENTATION IS NOTED AS 'S_F' SPECIAL CLASS FIBRE REINFORCED IT SHALL COMPLY WITH AS1379 AND HAVE THE FOLLOWING CONCRETE MIX PARAMETERS:

f'c 28 DAY COMPRESSIVE STRENGTH	REFER RELEVANT DRAWING
f'ct 56 DAY FLEXURAL STRENGTH	4.5MPa
SLUMP	SUPPLIER TO RECOMMEND TO SUIT PLACEMENT
AGGREGATE SIZE	20
PERMITTED ADMIXTURES	NONE
MINIMUM CEMENT RATIO	325kg/m^3
NOMINAL 56 DAY SHRINKAGE STRAIN	650um

5. ALL CONCRETE TO BE PLACED IN A CONTINUOUS OPERATION USING VIBRATING SCREED AND HANDHELD COMPACTORS TO ACHIEVE A DENSE, HOMOGENEOUS, VOID FREE SLAB.
6. CURING METHOD TO SUIT ARCHITECTURAL FINISHES AND BE COMPLIANT WITH CURRENT AUSTRALIAN STANDARDS AND MANUFACTURER'S SPECIFICATIONS. CURE FOR A MINIMUM OF 7 DAYS AFTER POUR.
7. CONCRETE TO BE PLACED CONTINUOUSLY BETWEEN CONSTRUCTION JOINTS SHOWN ON PLAN. DO NOT BREAK OR INTERRUPT POURS SUCH THAT COLD JOINTS OCCUR.
8. PROVIDE 2N16 TRIMMERS TOP AND BOTTOM TO ALL RE-ENTRANT CORNERS, AND PENETRATIONS IN SLAB. TRIMMERS TO EXTEND MINIMUM OF 600 PAST PENETRATION OR RE-ENTRANT CORNER.
9. THICKNESS ON STRUCTURAL DRAWINGS ARE MINIMUM. APPLIED FLOOR FINISHES, SET DOWNS, RAMPS AND FALLS DENOTED ON ARCHITECTURAL DRAWING ARE IN ADDITION TO MINIMUM STRUCTURAL THICKNESS UNO
10. SERVICE PIPES AND CONDUITS SHALL BE CHAIRED IN THE MIDDLE THIRD OF THE SLAB IN SUCH A MANOR THAT THEY DO NOT MOVE DURING THE CONCRETE POUR. THE MAXIMUM CAST IN SERVICE SIZE SHALL NOT EXCEED ONE FIFTH OF THE SLAB THICKNESS. ALL CAST IN SERVICES SHALL BE CONFIRMED ON PLANS WITH NORTHROP PRIOR TO CASTING INTO SLABS. SERVICES SHALL NOT TOUCH OR BE TIED TO SLAB REINFORCEMENT.
11. REFER ARCHITECT FOR ALL FALLS, STEPS, RAMPS, LEVELS.
12. BEAM SIZES ON PLAN ARE DENOTED DEPTH BY WIDTH.
13. NO HOLES CHASES OR EMBEDMENT OF PIPES OTHER THAN SHOWN ON STRUCTURAL DRAWINGS SHALL BE MADE TO CONCRETE MEMBERS WITHOUT WRITTEN APPROVAL OF NORTHROP CONSULTING ENGINEERS.
14. PROVIDE DRIP GROOVES AT ALL EXPOSED EDGES. CHAMFERS, DRIP GROOVES REGLETS TO ARCHITECTS DETAILS.
15. REFER ARCHITECT FOR CONCRETE SURFACE FINISHES. MINIMUM REQUIREMENTS AS FOLLOWS:

ELEMENT	SURFACE FINISH
CONCRETE SLABS U.N.O.	MACHINE FLOAT
TILED CONCRETE SLABS	WOOD FLOAT
FOOTPATHS	BROOM FINISH WITH ABRASIVE SLIP TESTING
COLUMNS AND WALLS	OFF FORM
STAIRS	STEEL TROWEL

CONCRETE MATERIALS

1. THE CONSTITUENTS OF CONCRETE SHALL BE IN ACCORDANCE WITH THE FOLLOWING:
   A. PORTLAND CEMENT TO BE GENERAL PURPOSE IN ACCORDANCE WITH AS3972.
   B. BLENDED CEMENTS ARE ONLY TO BE USED WHEN SPECIFIED IN ACCORDANCE WITH AS3972
   C. FLY ASH TO BE FINE GRADE ONLY IN ACCORDANCE WITH AS3582.1 AND ONLY USED WHEN SPECIFIED. AN ALTERNATE MIX DESIGN USING FLY ASH WILL ONLY BE CONSIDERED PROVIDED THE CONTENTIOUS MATERIAL MEETS THE PERFORMANCE REQUIREMENT OF THE CEMENT TYPE SPECIFIED.
   D. SILICA FUME ONLY TO BE USED WHEN SPECIFIED IN ACCORDANCE WITH AS972. AN ALTERNATE MIX DESIGN USING SILICA FUME WILL ONLY BE CONSIDERED PROVIDED THE CONTENTIOUS MATERIAL MEETS THE PERFORMANCE REQUIREMENT OF THE CEMENT TYPE SPECIFIED.
   E. AGGREGATE TO COMPLY WITH AS2758.1
      - FINE: DENSE NATURALLY OCCURRING SAND OR ROCK, CRUSHED OR UNCRUSHED, SINGLE SOURCE OR BLENDED CONFORMING WITH TABLE 3 OF AS2758.1
      - COURSE: CLEAN HARD DURABLE PARTICLES OF DENSE NATURALLY OCCURRING GRAVEL OR ROCK, CRUSHED OR NOT, EITHER SINGLE SOURCE OR BLENDED CONFORMING WITH GRADING REQUIREMENTS OF TABLE 1 - 20mm AGGREGATE OF AS2758.1
      PARTIAL DENSITY MINIMUM 2100kg/m^3 IN ACCORDANCE WITH CLAUSE 8.1 AS2758.1
      BULK DENSITY MINIMUM 2100kg/m^3 IN ACCORDANCE WITH CLAUSE 8.3 AS2758.1
      WATER ABSORPTION 2.5% IN ACCORDANCE WITH CLAUSE 8.3 AS2758.1
      DURABILITY "SEVERE" IN ACCORDANCE WITH CLAUSE 10 CONCRETE EXPOSURE CLASSIFICATION AS2758.1
      ALKALI REACTIVITY TESTING REQUIRED IN ACCORDANCE WITH CLAUSE 14.3 PROVIDE SEPARATE TESTING FOR EACH SINGLE SOURCE MATERIAL.
      PARTICLE SHAPE IN ACCORDANCE WITH CLAUSE 9.3 AND THE PROPORTION OF MISSHAPEN PARTICLES USING A 2:1 RATIO IS NOT TO EXCEED 35% WHEN DETERMINED IN ACCORDANCE WITH AS1141.14
   F. RECYCLED AGGREGATE TO BE CLASS 1 RCA IN ACCORDANCE WITH HB155 - GUIDE TO THE USE OF RECYCLED CONCRETE AND MASONRY MATERIALS. CONCRETE SUPPLIER TO PROVIDE ADDITIONAL M.O.R. AND COMPRESSIVE TESTING TO EACH BATCH TO CONFIRM RECYCLED AGGREGATE MIXES COMPLY WITH STRUCTURAL REQUIREMENTS. PERCENTAGE OF RECYCLED MATERIALS TO BE CONFIRMED WITH NORTHROP.
   G. WATER, FOR 'S' SPECIAL MIX CONCRETE WATER TO BE FROM POTABLE TOWN WATER SUPPLY. FOR 'N' NOMINAL CLASS CONCRETE RECYCLED WATER CAN BE USED
2. CONCRETE MIX DESIGNS SHALL BE PREPARED BY SPECIALIST CONCRETE TECHNOLOGIST AND SUBMITTED TO NORTHROP CONSULTING FOR APPROVAL PRIOR TO PRODUCTION OF CONCRETE FOR USE ON SITE.

CONCRETE NOTES CONTINUED

CONCRETE SUPPLY AND DELIVERY

1. THE SUPPLY OF CONCRETE SHALL BE FROM A CONCRETE BATCHING PLANT THAT CARRIES OUT PRODUCTION ASSESSMENT FOR THE GRADES OF CONCRETE SPECIFIED BY THIS PROJECT.
2. WATER IS NOT TO BE ADDED TO MIX AFTER BATCHING UNLESS SPECIFIED BY MIX DESIGN ENGINEER.
3. ELAPSED TIME BETWEEN THE WETTING OF THE MIX AND THE DISCHARGE OF THE MIX AT SITE MUST NOT EXCEED THE FOLLOWING CRITERIA:

CONCRETE TEMPERATURE AT TIME OF DISCHARGE	MAXIMUM ELAPSED TIME IN MINUTES
5°c TO 24°c	120
24°c TO 27°c	90
27°c TO 30°c	60
30°c TO 32°c	45
32°c TO 35°c	30

   IF THE ELAPSED DELIVERY TIME IS GREATER THAN THE TIMES ABOVE OR THE CONCRETE TEMPERATURE IS OUTSIDE THE TEMPERATURES SPECIFIED THE CONCRETE IS TO BE REJECTED.
4. IF A POUR IS STOPPED, PRIOR TO FURTHER PLACEMENT OF CONCRETE, NORTHROP ENGINEERS ARE TO BE CONTACTED TO DETERMINE WHAT, IF ANY RECTIFICATION WORKS ARE REQUIRED.
5. IF AIR TEMPERATURE IS PREDICTED AT BEING LESS OR EQUAL TO 10°c FOR A 12 HOUR PERIOD "COLD WEATHER" CONCRETING PROCEDURES ARE TO BE ADOPTED. REFER MIX DESIGN ENGINEER AND NORTHROP FOR DETAILS.

CONCRETE TESTING

1. THE ORGANIZATION RESPONSIBLE FOR SAMPLING AND TESTING CONCRETE ARE TO HAVE RELEVANT 'NATA' LABORATORY REGISTRATION, BE INDEPENDENT, AND USE TRAINED, COMPETENT PERSONNEL FOR THE TAKING OF SAMPLES AND SPECIMENS AND THE PREPARATION OF MATERIALS AND WORK FOR TESTING.
2. <u>SLUMP TESTING</u>: ALL CONCRETE AT TIME OF PLACEMENT IS TO HAVE A SLUMP TEST TO BE WITHIN THE PERMISSIBLE TOLERANCE AS SPECIFIED IN AS1379 FOR THE NOMINAL SUMP SPECIFIED IN THE MIX DESIGN.
3. <u>PRODUCTION ASSESSMENT TESTING</u>: FOR 'N' CLASS CONCRETE BEING PLACED ON GROUND WITH A SPECIFIED THICKNESS LESS THAN OR EQUAL TO 110, PRODUCTION ASSESSMENT TESTING IN ACCORDANCE WITH AS1379, CARRIED OUT BY THE BATCHING PLANT IS ACCEPTABLE. EVIDENCE OF PRODUCTION TESTING TO BE PROVIDED PRIOR TO POUR.
4. <u>PROJECT ASSESSMENT TESTING</u>: ALL CONCRETE (EXCLUDING PRODUCTION ASSESSMENT SLABS) SHALL HAVE TESTING SAMPLES COLLECTED AT DISCHARGE POINT IN ACCORDANCE WITH AS1379 AND THE FOLLOWING SPECIFICATION:
   A. A SAMPLE COMPRISES OF:

CONCRETE CLASSIFICATION	No. CYLINDERS PER SAMPLE	TIME OF, AND PERCENTAGE STRENGTH REQUIRED AT TESTING				
		24 HOURS f'c	4 DAY f'c	7 DAY f'c	28 DAY f'c	56 DAY f'ct
N	3	NIL	NIL	1 CYL 50%	2 CYL/100%	NIL
S	3	NIL	NIL	1 CYL 50%	2 CYL/100%	NIL
S_F	4	NIL	NIL	1 CYL 50%	2 CYL/100%	1 CYL 100%
S_P	5	1 CYL 10MPa	1 CYL 22MPa	1 CYL 75%	2 CYL/100%	NIL

   ADDITIONAL CYLINDERS CAN BE COLLECTED AND TESTED AT CONTRACTORS DISCRETION.
   CYLINDERS ARE TO BE MADE IN ACCORDANCE WITH AS1012.8.1 AND AS1012.8.2
   B. THE NUMBER OF SAMPLES TO BE COLLECTED ARE TO COMPLY WITH THE FOLLOWING TABLE:

VOLUME OF CONCRETE BEING PLACED	NUMBER OF TEST SAMPLES
UP TO 10m^3	1
10m^3 TO 30m^3	2
30m^3 TO 50m^3	3
OVER 50m^3	1 SAMPLE FOR EVERY 50m^3 MIN OF 3 SAMPLES

   C. IF MULTIPLE CONCRETE GRADES ARE BEING POURED THE VOLUME OF EACH GRADE SHOULD BE CONSIDERED SEPARATELY FOR PURPOSE OF CALCULATING NUMBER OF TEST SAMPLES TO BE COLLECTED.
   D. WHEN MULTIPLE SAMPLES ARE BEING COLLECTED. THE FIRST AND LAST CONCRETE BATCH WILL BE SAMPLED AND OTHER CONCRETE BATCHES WILL BE SAMPLED PROGRESSIVELY THROUGH OUT THE POUR.
   E. SAMPLES COLLECTED ARE TO BE TESTED; FOR 'N' CLASS CONCRETE 1 CYLINDER AT 7 DAYS AND 2 CYLINDERS AT 28 DAYS. FOR 'S' CLASS CONCRETE; 1 CYLINDER AT 1 DAY (10MPa), 1 CYLINDER AT 4 DAYS (22MPa) 1 CYLINDER AT 7 DAYS AND 2 AT 28 DAYS.
5. ALL STEEL FIBRE REINFORCED CONCRETE (SFRC) TO BE TESTED FOR FIBRE CONTENT IN ACCORDANCE WITH AS3600 CLAUSE 16.7.5

REINFORCEMENT

1. ALL REINFORCING BARS SHALL BE GRADE D500N TO AS/NZS 4671 UNLESS NOTED OTHERWISE.
2. ALL MESH SHALL BE GRADE 500L TO AS/NZS 4671 UNLESS NOTED OTHERWISE
3. THE FOLLOWING TABLE DENOTES REINFORCEMENT GRADES AND NOTATIONS USED IN DOCUMENTATION

SYMBOL	BAR DESCRIPTION	GRADE MPa	DUCTILITY CLASS	AS/ NZ STD
S	STRUCTURAL GRADE DEFORMED BAR	250N	NORMAL	4671
N	HOT ROLLED DEFORMED RIB BAR	500N	NORMAL	4671
R	ROUND BAR	250N	NORMAL	4671
RL	RECTANGULAR MESH DEFORMED RIB BAR	500L	LOW	4671
SL	SQUARE MESH DEFORMED RIB BAR	500L	LOW	4671
TM	TRENCH MESH	500L	LOW	4671

4. EXCEPT FOR MANUFACTURED MESH, CLASS L REINFORCEMENT SHALL NOT BE USED.
5. REINFORCEMENT IS REPRESENTED DIAGRAMMATICALLY, AND NOT NECESSARILY IN TRUE PROJECTION. BARS SHOWN ARE INDICATIVE ONLY AND LENGTHS MAY VARY. REINFORCEMENT SCHEDULER TO SIZE BARS BASED ON ARCHITECTURAL SLAB SET OUT MAKING ALLOWANCES FOR LAPS COGS AND EMBEDMENT.
6. REINFORCEMENT IS SPECIFIED ACROSS PLANS ELEVATIONS AND SECTIONS. REFER ALL ELEMENTS FOR COMPLETE REINFORCING REQUIREMENTS.
7. USE ONLY PLASTIC OR CONCRETE CHAIRS AT EXTERNAL SURFACES.
8. SITE BENDING OF REINFORCEMENT BARS SHALL BE DONE WITHOUT HEATING USING A RE-BENDING TOOL. THE BARS SHALL BE RE-BENT AGAINST A PIN WITH A DIAMETER NOT LESS THAN THE MINIMUM PIN SIZE PRESCRIBED IN AS3600.
9. SPLICES IN REINFORCEMENT SHALL BE MADE ONLY IN POSITIONS SHOWN ON THE STRUCTURAL DRAWINGS OR IN POSITIONS OTHERWISE APPROVED IN WRITING BY NORTHROP CONSULTING ENGINEERS. LAPS SHALL NOT BE LESS THAN THE DEVELOPMENT LENGTH FOR EACH BAR AND IN ACCORDANCE WITH AS3600 SECTION 13.
10. WELDING OF REINFORCEMENT SHALL NOT BE PERMITTED UNLESS SHOWN ON THE STRUCTURAL DRAWINGS OR APPROVED BY NORTHROP CONSULTING ENGINEERS.
11. AT EXTERNALLY EXPOSED SURFACES NO METALLIC ITEMS INCLUDING FORM BOLTS, FORM SPACERS, METALLIC BAR CHAIRS AND TIE-WIRE ARE TO BE PLACED IN THE COVER ZONE.
12. ALL REINFORCEMENT, ANCHOR BOLTS AND OTHER CONCRETE INSERTS SHALL BE WELL SECURED IN POSITION AND INSPECTED BY NORTHROP CONSULTING ENGINEERS PRIOR TO PLACING CONCRETE.
13. HOLES FOR POST FIXED REINFORCEMENT SHALL BE DRILLED WITH HAMMER DRILL TO THE REQUIRED DEPTH. CORE HOLES SHALL NOT BE USED.
14. CHEMICAL ADHESIVE TO BE EITHER RAMSET REO 502 OR HILTI RE-500V3 STRICTLY INSTALLED TO MANUFACTURERS SPECIFICATIONS.
15. ALL STARTER BARS AND REINFORCEMENT BEING PLACED THROUGH POUR BREAKS ARE TO BE SUITABLY TIED IN POSITION SO THEY DO NOT MOVE. 'WET SETTING' OF BARS IS NOT PERMITTED.
16. ALL REINFORCEMENT IS TO BE INSPECTED BY NORTHROP AFTER IT IS TIED IN POSITION AND PRIOR TO CONCRETE BEING POURED. CONTRACTOR TO ALLOW ENOUGH TIME FOR CORRECTIONS OF ANY ERRORS IN PLACEMENT PRIOR TO POUR.

CONCRETE NOTES CONTINUED

FORMWORK

1. THE DESIGN, CERTIFICATION, CONSTRUCTION, INSPECTION AND PERFORMANCE OF THE FORMWORK AND FALSE WORK SHALL BE THE RESPONSIBILITY OF THE FORMWORK SUB-CONTRACTOR, EXCEPT TO THE EXTENT THAT FORMWORK DESIGN IS SHOWN ON THE STRUCTURAL DRAWINGS.
2. FORMWORK SHALL BE CERTIFIED BY A STRUCTURAL ENGINEER EXPERIENCED IN FORMWORK DESIGN IN ACCORDANCE WITH WORKCOVER REGULATIONS AND THE WORKCOVER CODE OF PRACTICE.
3. FORMWORK SHALL BE DESIGNED IN ACCORDANCE WITH AS3610-1995. THE DESIGN SHALL ACCOMMODATE MOVEMENTS AND LOAD RE-DISTRIBUTION DUE TO ANY POST TENSIONING.
4. PROVIDE RESTRAINT OR SUPPORT TO ENSURE STABILITY OF FORMWORK THAT IS INDEPENDENT OF THE PERMANENT STRUCTURE. APPROVAL FROM NORTHROP CONSULTING ENGINEERS IS REQUIRED IF FORMWORK SUPPORT IS REQUIRED FROM THE PERMANENT STRUCTURE.
5. FOUNDATIONS SUPPORTING THE FORMWORK SHALL BE DETERMINED BY THE FORMWORK SUB-CONTRACTOR FROM THE CONDITIONS EXISTING ON SITE AT THE TIME OF CONSTRUCTION. REFER TO THE GEOTECHNICAL REPORT FOR THE SITE.
6. FORMWORK CONSTRUCTION DIMENSIONAL TOLERANCES AND STRIPPING TIMES SHALL COMPLY WITH AS3610-1995 AND AS3600-2018 UNLESS OTHERWISE APPROVED BY NORTHROP CONSULTING ENGINEERS.
7. DURING CONSTRUCTION, SUPPORT PROPPING WILL BE REQUIRED WHERE LOADS FROM STACKED MATERIALS, FORMWORK AND OTHER SUPPORTED SLABS INDUCE LOADS IN A SLAB OR BEAM WHICH EXCEED THE DESIGN CAPACITY FOR STRENGTH OR SERVICEABILITY LIMIT STATES AT THAT AGE. ONCE THE NOMINATED 28 DAY STRENGTH HAS BEEN ATTAINED, THESE LOADS SHALL NOT EXCEED THE DESIGN SUPERIMPOSED LOADS SET OUT IN DESIGN LOADS TABLE ON THE RELEVANT DRAWING.
8. STRIPPING OF FORMWORK AND OR REMOVAL OF ANY PROPPING SHALL NOT BE CONDUCTED PRIOR TO SEVEN DAYS AFTER SLAB POUR NOT COUNTING THE DAY OF POUR.
9. THE FORMWORK FINISH SHALL COMPLY WITH AS3610.1 CLASSIFICATIONS AS SPECIFIED IN THE FOLLOWING TABLE:

ELEMENT	CLASS	ELEMENT	CLASS
FOOTINGS	4	EXPOSED SUSPENDED SLABS	2
EXPOSED SLAB ON GROUND	2	CONCEALED SUSPENDED SLABS	3
CONCEALED SLAB ON GROUND	3	SUSPENDED SLABS SOFFITS	3
COLUMNS	2	STAIR LANDINGS AND FLIGHTS	2
WALLS	2	STAIR AND ELEVATOR SHAFT LIDS	3

FOOTING NOTES

GENERAL

1. A GEOTECHNICAL REPORT HAS BEEN CARRIED OUT AND IS AVAILABLE FOR VIEWING AT NORTHROP CONSULTING. REPORT No. **229444.01** REV **0** DATED **4/09/2025** PREPARED BY DOUGLAS PARTNERS PTY LTD. THIS REPORT IS FOR INFORMATION ONLY, IT IS NOT A COMPLETE DESCRIPTION OF CONDITIONS AT OR BELOW GROUND LEVEL.
2. THE CONTRACTOR SHALL ALLOW TO ENGAGE A QUALIFIED RPEQ GEOTECHNICAL ENGINEER TO APPROVE THE FOUNDATION MATERIAL. OBTAIN GEOTECHNICAL ENGINEERS APPROVAL AND SUBMIT CERTIFICATE IN WRITING TO NORTHROP CONSULTING ENGINEERS PRIOR TO CONCRETING FOUNDATIONS.
3. FOOTINGS HAVE BEEN DESIGNED USING THE FOLLOWING ASSUMED CAPACITIES:

ASSUMED ALLOWABLE BEARING CAPACITY UNLESS NOTED OTHERWISE	
PAD FOOTINGS	150 kPa ALLOWABLE
STRIP FOOTINGS	150 kPa ALLOWABLE

   INCREASE DESIGN FOOTING DEPTH WITH 10MPa MASS CONCRETE AS REQUIRED TO ACHIEVE BEARING CAPACITY
4. TOP SOIL AND DETRITUS MATTER IS TO BE REMOVED FROM BUILDING FOOTPRINT IN ACCORDANCE WITH GEOTECHNICAL ENGINEERS INSTRUCTIONS.
5. THE BUILDING PLATFORM SHALL BE PROOF ROLLED WITH 6 PASSES A 10 TONNE VEHICLE UNDER THE SUPERVISION OF CIVIL ENGINEER AND OR GEOTECHNICAL ENGINEER. IDENTIFIED SOFT AREAS ARE TO BE CORRECTED.
6. FOOTINGS ARE TO FOUND ON THE SAME EARTH STRATA. INCREASE FOOTING DEPTH WITH 10MPa CONCRETE TO OBTAIN SIMILAR STRATA.
7. THE BUILDING PLATFORM IS TO BE PREPARED UNDER LEVEL 1 GEOTECHNICAL SUPERVISION. REFER CIVIL ENGINEER AND GEOTECHNICAL ENGINEER FOR PREPARATION DETAILS.
8. WHEN SERVICES ARE ADJACENT FOOTINGS AND WITHIN A 45° LINE OF INFLUENCE FROM BASE OF FOOTING INCREASE FOOTING DEPTH WITH 10MPa MASS CONCRETE SO THAT SERVICE INCLUDING TRENCH IS OUTSIDE THE LINE OF INFLUENCE.
9. ALL SERVICES ARE TO CROSS FOOTINGS PERPENDICULAR TO FOOTING. BACKFILL UNDER FOOTING WITH 10MPa MASS CONCRETE PROVIDE MOVEMENT JOINTS AS REQUIRED TO SERVICE TO SERVICE ENGINEERS DETAIL.
10. SERVICES ARE TO CROSS PERPENDICULAR TO STRIP FOOTINGS. REFER SERVICE CONSULTANT FOR APPLICABLE MOVEMENT JOINTS AT MASS CONCRETE ENCASEMENTS.
11. DO NOT ALLOW EXCAVATED MATERIAL TO BE STOCKPILED WITHIN 1500mm OF FOOTING TRENCHES OR PITS. NO EARTH OR DETRITUS IS TO FALL INTO THE FOOTING TRENCHES BEFORE OR DURING CONCRETE PLACEMENT.
12. FOOTINGS SHALL BE EXCAVATED TO THE DETAILED DEPTH AND WIDTH. FOOTINGS SHALL BE INSPECTED AND FILLED WITH CONCRETE AS SOON AS POSSIBLE TO AVOID EITHER SOFTENING OF THE FOUNDATION.
13. THE BASE OF ALL PIER HOLES SHALL BE FREE OF WATER AND CLEANED OF LOOSE MATERIAL OR DEBRIS PRIOR TO PLACEMENT OF CONCRETE. ALLOW TO PROVIDE TEMPORARY LINERS AS DEEMED NECESSARY.

DRN	AMENDMENTS	DATE	REV
NS	BAC APPROVAL	11.06.26	B
NS	PRELIM STEEL ISSUE	29.05.26	A

NORTHROP
PROJECT NUMBER: 250946

Drawn: HP	Checked:
Design Coordinated: HP	Compliance Checked: –
Authorised For Issue: NS	Construction Checked: –

SKYGATE PLAY

STRUCTURAL SPECIFICATIONS SHEET 1

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Size: A1
Revision: B

Drawing Number	Project No.	Originator	Volume	Level	Type	Discipline	Number
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